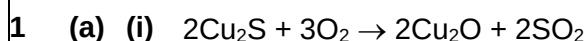


Worked solutions for 2017 9729 H2 Chemistry Preliminary Examination Paper 3



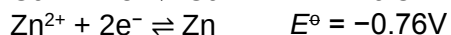
- (ii) Both compounds have giant ionic lattice structure. While the charge and size of the cation as well as the charge of the anion are the same in both ionic compounds, **S^{2-} is bigger than O^{2-}** .

Consequently, **the electrostatic forces of attraction between Cu^+ and S^{2-} are weaker than that between Cu^+ and O^{2-}** . Hence, **lesser amount of energy is required** to overcome the weaker ionic bonds in Cu_2S . The melting point of **Cu_2S will be lower than that of Cu_2O** .

[1]: correct comparison of the size of S^{2-} and O^{2-}

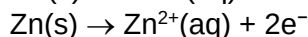
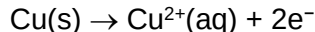
[1]: Cu_2S has a lower melting point than Cu_2O

- (iii) anode: impure copper
cathode: pure copper
electrolyte: $\text{CuSO}_4(\text{aq})$



reaction at the anode

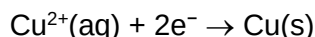
By controlling the voltage, when the current runs through the circuit, copper and zinc metals will be oxidised to form aqueous cations at the anode.



Ag will not be oxidised due to its more positive $E^\ominus$ reduction potential (compared to Cu), and will drop to the bottom as sludge.

reaction at the cathode

Subsequently both Cu^{2+} and Zn^{2+} will migrate to the cathode but only Cu^{2+} will be reduced to Cu as it has a more positive $E^\ominus$ reduction potential than Zn.



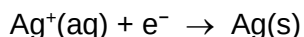
Hence, the copper cathode would grow in size with pure copper deposited, while Zn^{2+} remains in solution.

[1]: description of anode, cathode and electrolyte

[1]: correct electrode reactions (both anode and cathode) with reference to relevant $E^\ominus$ values (correct state symbols must be given for the equations)

[1]: Ag dropping to the bottom as sludge and Zn^{2+} remaining in the solution

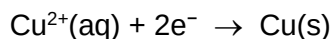
- (iv) Amount of Ag deposited = $0.100 \div 107.9 = 9.2678 \times 10^{-4} \text{ mol}$



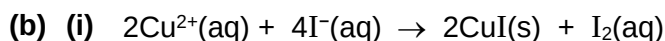
$$\text{Amount of electrons} = 9.2678 \times 10^{-4} \text{ mol}$$

$$Q = (9.2678 \times 10^{-4}) \times 96500 = 89.436 \text{ C}$$

$$I = 89.436 \div (30 \times 60) = 0.0497 \text{ A}$$



Amount of copper deposited = $\frac{1}{2} \times (9.2678 \times 10^{-4}) = 4.6339 \times 10^{-4} \text{ mol}$
 Mass of copper deposited = $(4.6339 \times 10^{-4}) \times 63.5 = 0.029425 = 0.0294 \text{ g}$

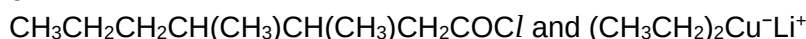
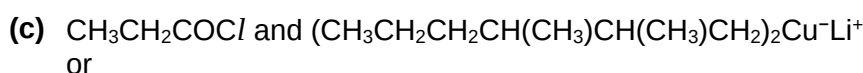


$$E^{\circ}_{\text{cell}} = +0.15 - (+0.54) = -0.39\text{V}$$

Since the E°_{cell} is **negative**, the reaction is not feasible.

(ii) The precipitation of CuI reduces the concentration $\text{Cu}^{+}(\text{aq})$, shifting the $\text{Cu}^{2+}/\text{Cu}^{+}$ equilibrium ($\text{Cu}^{2+} + \text{e}^{-} \rightleftharpoons \text{Cu}^{+}$) to the right, favouring the reduction process and hence making the reduction potential of $\text{Cu}^{2+}/\text{Cu}^{+}$ more positive.

As such, E°_{cell} becomes positive and the redox reaction becomes spontaneous.



[1] for a suitable Gilman reagent, and [1] for the corresponding acid chloride

(d) (i) Entropy (S) is a measure of the degree of disorder or randomness in a system.

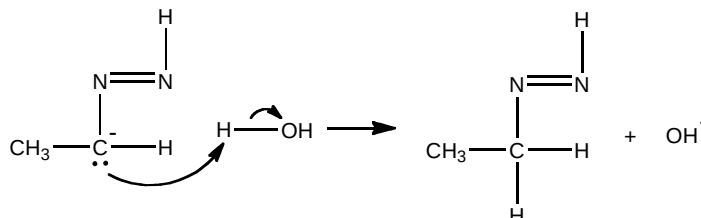
(ii) ΔS is **positive** as the forward reaction is **accompanied by an increase in the number of moles of gases**. Hence, there is an increase in the number of ways of distributing the gaseous particles and entropy increases.

(iii) $\Delta H^{\circ} = [(-84.7) + (-285.8)] - [(-196.4) + 50.6] = -224.7 \text{ kJ mol}^{-1}$

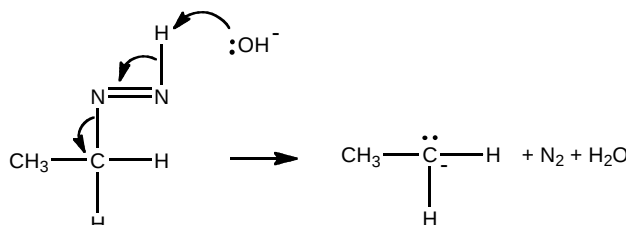
$$\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ} = -224.7 - (298)(0.1915) = -281.8 \text{ kJ mol}^{-1}$$

(iv) condensation (or addition-elimination) reaction

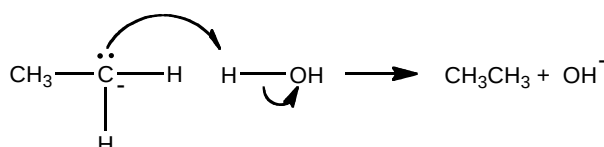
(v) step 3:



step 4:



step 5:



[Total: 21]

2 (a) (i) Le Chatelier's principle states that when a **system at equilibrium** is subjected to **external disturbance**, the **position of equilibrium will shift to counteract the change** imposed to re-establish the equilibrium.

(ii) As $[H^+]$ is increased, the **position of equilibrium shifts to the left to reduce $[H^+]$** , producing **higher concentration of $Cr_2O_7^{2-}$** and turning the solution orange.

(iii) The colour change implies that the **position of equilibrium has shifted to the right to absorb some of the added heat**.

Hence, the forward reaction is **endothermic**.

$$(iv) K_c = \frac{[CrO_4^{2-}]^2 [H^+]^2}{[Cr_2O_7^{2-}]}$$

$$= \frac{[CrO_4^{2-}]^2 (10^{-8})^2}{[Cr_2O_7^{2-}]} = 1.31 \times 10^{-13}$$

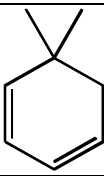
$$\frac{[CrO_4^{2-}]^2}{[Cr_2O_7^{2-}]} = 1310$$

(v) The high ratio of $\frac{[CrO_4^{2-}]^2}{[Cr_2O_7^{2-}]}$ indicates a much **higher proportion of CrO_4^{2-}** compared to $Cr_2O_7^{2-}$.

The solution will appear **yellow**.

(allow for ecf from (iii))

(b) (i)

	$(COOH)_2$
A	C

(ii) **A** does **not** exhibit **cis-trans isomerism** because only the *cis*-form can exist due to **ring strain**.

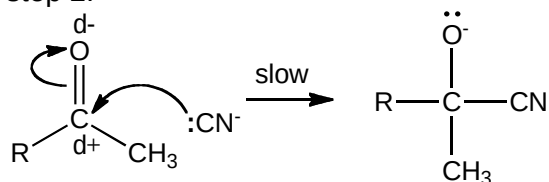
(iii) Compound **B** is a stronger acid than 3,3-dimethylbutanoic acid, because the conjugate base of compound **B** is **more stable** than the conjugate base of 3,3-dimethylbutanoic acid due to the **additional stabilisation of the hydrogen bond formed between the $-COO^-$ group of the mono-anion with the unionised $-COOH$ group**.

(c) nucleophilic addition

step 1:

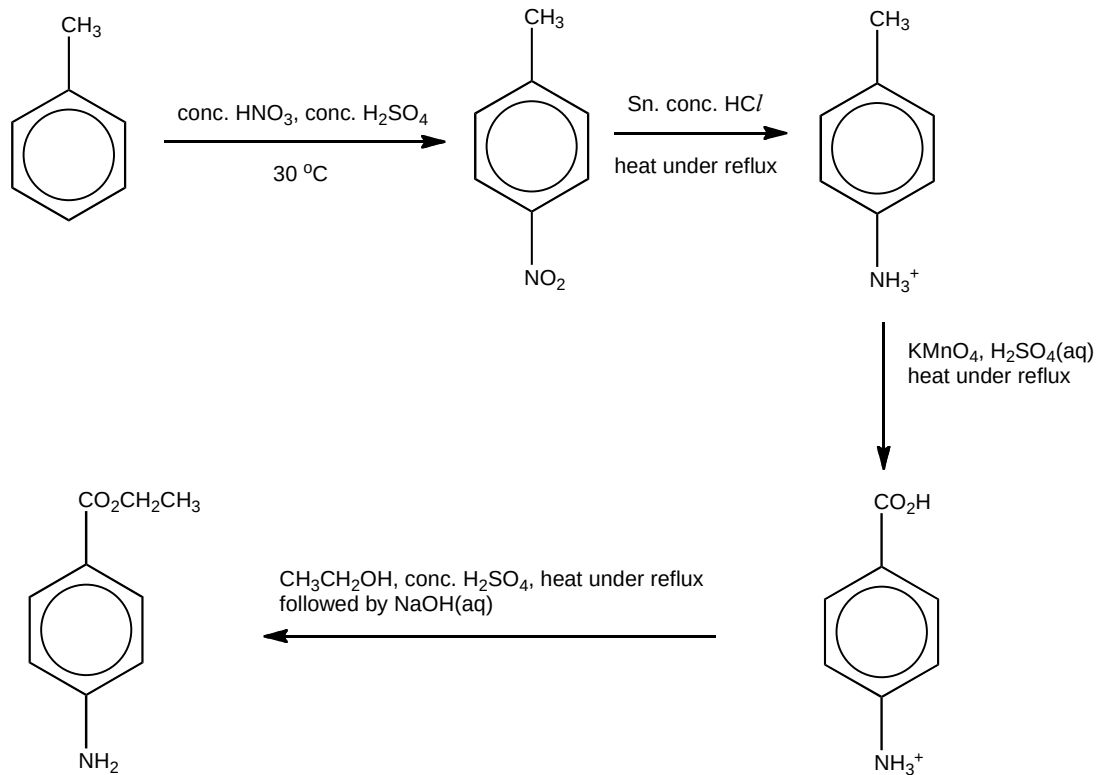
NaOH(aq) used	$\text{HCN} + \text{OH}^- \rightarrow \text{H}_2\text{O} + \text{CN}^-$
NaCN(s) used	$\text{NaCN(s)} + \text{aq} \rightarrow \text{Na}^+(\text{aq}) + \text{CN}^-(\text{aq})$

step 2:



step 3:

NaOH(aq) used	
NaCN(s) used	

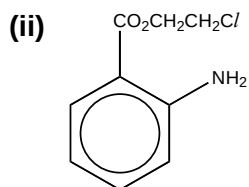
(d) (i)

- (ii) To a sample of the unknown in a test-tube, add **Br₂(aq)** until in excess.

If the sample is benzocaine, **orange Br₂** will be **decolourised** and a **white precipitate** is formed.

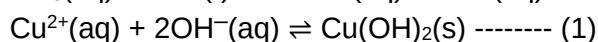
If the sample is methylbenzene, orange Br₂ will **not be decolourised** and **no white precipitate** is formed.

- (e) (i) (limited) Cl₂, uv



[Total: 21]

- 3 (a) (i) $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$



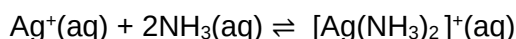
NH₃(aq) undergoes **hydrolysis in water** to form OH⁻. When a **small amount of NH₃** is added, a **blue precipitate** of **Cu(OH)₂** is formed.



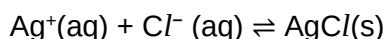
Addition of **excess ammonia** causes the formation of the more stable **deep blue [Cu(NH₃)₄(H₂O)₂]²⁺** complex ion through a **ligand-exchange reaction**.

The reaction **lowers the concentration of Cu²⁺(aq)** causing Cu(OH)₂ to dissolve. The **position of equilibrium (1) shifts to the left** and the **ionic product < K_{sp}**.

- (ii) When NH₃(aq) is added, it reacts with Ag⁺ to form [Ag(NH₃)₂]⁺(aq).



The **concentration of Ag⁺(aq) decreases** and the **equilibrium position shifts left** and **ionic product [Ag⁺][Cl⁻] becomes lower than K_{sp} (AgCl)**

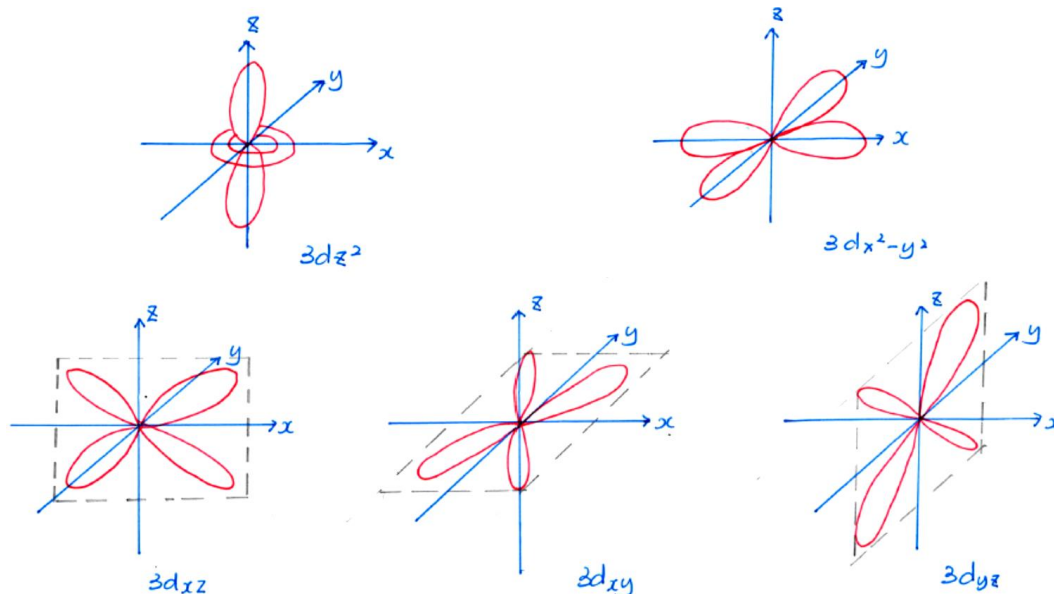


As the **K_{sp} of AgBr is lower than K_{sp} of AgCl**, the **ionic product [Ag⁺][Br⁻] remains higher than K_{sp} of AgBr** upon addition of NH₃(aq). Hence, the precipitate remains insoluble.

- (b) (i) The **energy difference between the 3d and 4s subshells in Fe is relatively small**.

This means with sufficient energy, **both the 4s and some / all of the inner 3d electrons can be removed** to form stable compounds and ions of various oxidation states.

(ii) $\text{Fe}^{3+}: 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$



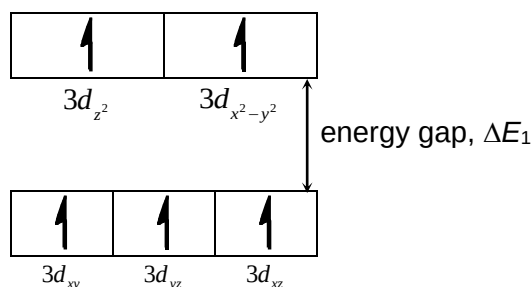
(iii) As the lobes of the $3d_{x^2-y^2}$ and $3d_{z^2}$ orbitals lie along the axes, the electrons in these two d-orbitals are pointing towards the lone pairs of electrons of the ligands in an octahedral complex. Repulsion between the lone pair of electrons of the ligands and the $3d$ electrons in these orbitals is stronger. Thus, $3d_{x^2-y^2}$ and $3d_{z^2}$ orbitals form the higher energy set.

As the $3d$ orbitals in Fe^{3+} are **partially filled**, d-d transition, where the **electron is promoted from a lower energy orbital to a higher energy orbital**, is possible.

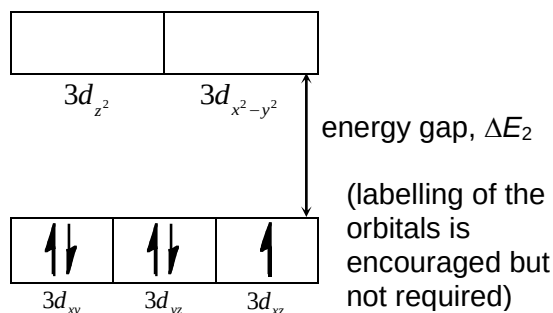
Energy that corresponds to the frequency of light in the visible region of the electromagnetic spectrum is absorbed. Colours that are complementary to those frequency of light absorbed are seen.

(iv) The **red complex absorbs visible light in the blue region** while the blue complex absorbs in the red region, i.e. the red complex absorbs light with a shorter wavelength, and so **the red complex has a larger energy gap** between the d-orbitals.

(v) high spin state for the blue complex



low spin state for the red complex



- [1] for each correctly drawn spin state
- [1] for indicating the energy gap size in the diagram such that $\Delta E_1 < \Delta E_2$

[Total: 18]

- 4 (a) (i) Benzaldehyde is simultaneously oxidised to benzoic acid and reduced to phenylmethanol.

(Or carbon is simultaneously oxidised as its oxidation state is increased from +1 in benzaldehyde to +3 in benzoic acid and reduced, as its oxidation state is decreased from +1 in benzaldehyde to -1 in phenylmethanol.)

- (ii) Comparing experiments 1 and 2, concentration of $\text{C}_6\text{H}_5\text{CHO}$ is kept constant, when concentration of KOH doubles, the rate doubles. Hence order of reaction with respect to KOH is 1.

Comparing experiments 2 and 3, as the order w.r.t. KOH is one, when the concentration of KOH is doubled, the rate should be doubled. However, when both the concentration of KOH and $\text{C}_6\text{H}_5\text{CHO}$ is doubled, the rate increased by 8 times. This means that when the concentration of $\text{C}_6\text{H}_5\text{CHO}$ is doubled, the rate is quadrupled. Hence order of reaction with respect to $\text{C}_6\text{H}_5\text{CHO}$ is 2.

- (iii) $\text{rate} = k [\text{C}_6\text{H}_5\text{CHO}]^2 [\text{KOH}]$

For experiment 1: $k = 0.0127 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$

For experiment 2: $k = 0.0122 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$

For experiment 3: $k = 0.0128 \text{ mol}^{-2} \text{ dm}^6 \text{ s}^{-1}$

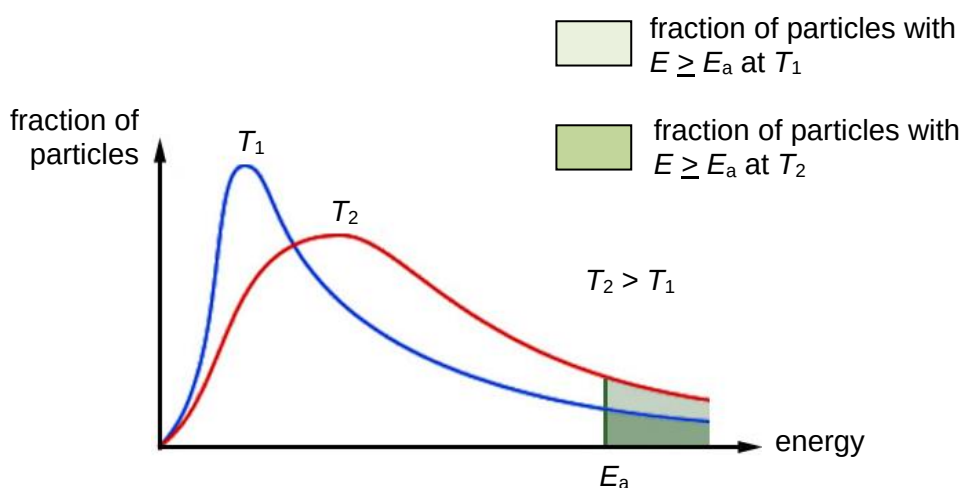
- [1] for any one out of the three

- (iv) $\text{rate} = k' [\text{KOH}]$ where $k' = k[\text{C}_6\text{H}_5\text{CHO}]^2$

$$t_{1/2} = \frac{\ln 2}{k[\text{C}_6\text{H}_5\text{CHO}]^2} = \frac{\ln 2}{(0.012667)(1.60)^2} = 21.4 \text{ s}$$

- (v) When temperature increases, **molecular speed and average kinetic energies of reactant particles increases**. Hence, there will be a **higher frequency of collisions** and a **greater fraction** of the particles will have energy **greater than the activation energy** required for reaction. This will lead to a **higher frequency of effective collisions**.

The **rate constant k increases** as the temperature increases, hence **rate of reaction increases**.

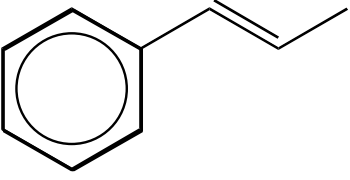
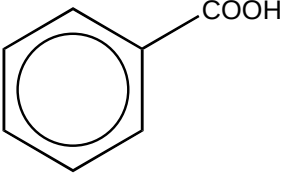
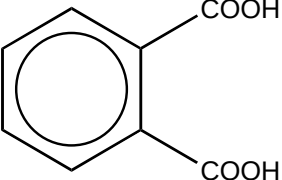


- (b) (i) LiAlH_4 in dry ether / NaBH_4 in methanol, room temperature

- (ii) KMnO_4 / $\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4 , reflux

(c)	test	benzaldehyde	phenylmethanol
	anhydrous PCl_5 / SOCl_2	no white fumes	white fumes
	Tollens' reagent, warm	silver mirror	no silver mirror
	2,4-DNPH	orange precipitate	no orange precipitate
	Na	no effervescence	effervescence and the gas evolved 'pops' with a lighted splint

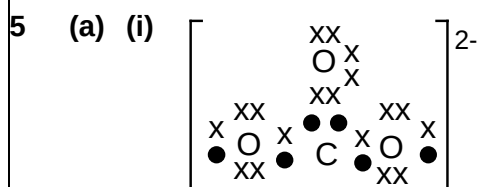
[1] for correct reagents and conditions, and [1] for correct observations for both benzaldehyde and phenylmethanol

(d)			
	R	T	U

[1] for each correct structure

information	deduction
R exhibits stereoisomerism.	R exhibits cis-trans isomerism (it cannot exhibit enantiomerism as the side chain contains only three C-atoms). R contains C=C group (with two different groups attached to each C-atom)
R is found to decolourise orange aqueous bromine but S does not.	R undergoes electrophilic addition with aqueous bromine. R contains C=C group
When hot acidified potassium manganate solution is added to 1 mol of R and S , both decolourised the purple potassium manganate to form compound T and U respectively.	R and S undergoes oxidation Presence of benzylic H on both R and S both T and U contains -COOH group
1.0 mol of compound T exactly reacts with 0.5 mol of sodium carbonate while compound 1.0 mol of compound U reacts with exactly 1.0 mol of sodium carbonate to give out carbon dioxide gas.	T and U undergoes neutralisation reaction with sodium carbonate. T contains one -COOH group U contains two -COOH groups

[Total: 20]



(ii) trigonal planar

the **3 bond pairs and 0 lone pair** of electrons **will space themselves out as far apart as possible to minimise repulsion.**

(b) (i) The reaction between Na_2CO_3 and HCl is an acid-base reaction as HCl functions as a **Bronsted acid** and **donates a proton** to Na_2CO_3 . Hence, Na_2CO_3 functions as a **Bronsted base** and **accepts a proton** from HCl .

(ii) $\text{HCO}_3^- + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 + \text{OH}^-$

The pH is weakly basic as HCO_3^- undergoes hydrolysis in water to produce OH^- .

(c) When a **small amount of acid** is added,
 $\text{HCO}_3^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq}) \rightarrow \text{H}_2\text{CO}_3(\text{aq}) + \text{H}_2\text{O}(\text{l})$

When a **small amount of alkali** is added:

$\text{H}_2\text{CO}_3(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l})$

(d) (i) Ideal gas particles take up zero (or negligible) volume (as compared to the volume of the container)

There are no (or negligible) intermolecular forces of attraction between the particles of an ideal gas.

(ii) $pV = nRT$

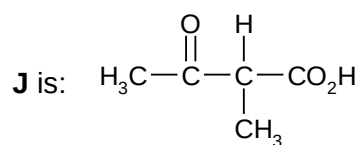
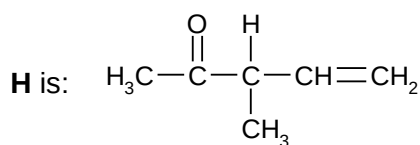
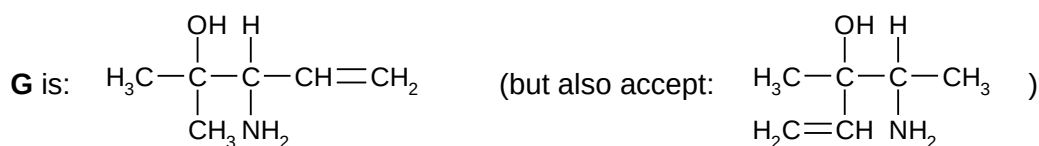
$$1.1 \times 101325 \times 56.0 \times 10^{-6} = n \times 8.31 \times (30 + 273)$$

$$\text{No. of moles of } \text{CO}_2 \text{ produced} = \mathbf{2.4789 \times 10^{-3} \text{ mol}}$$

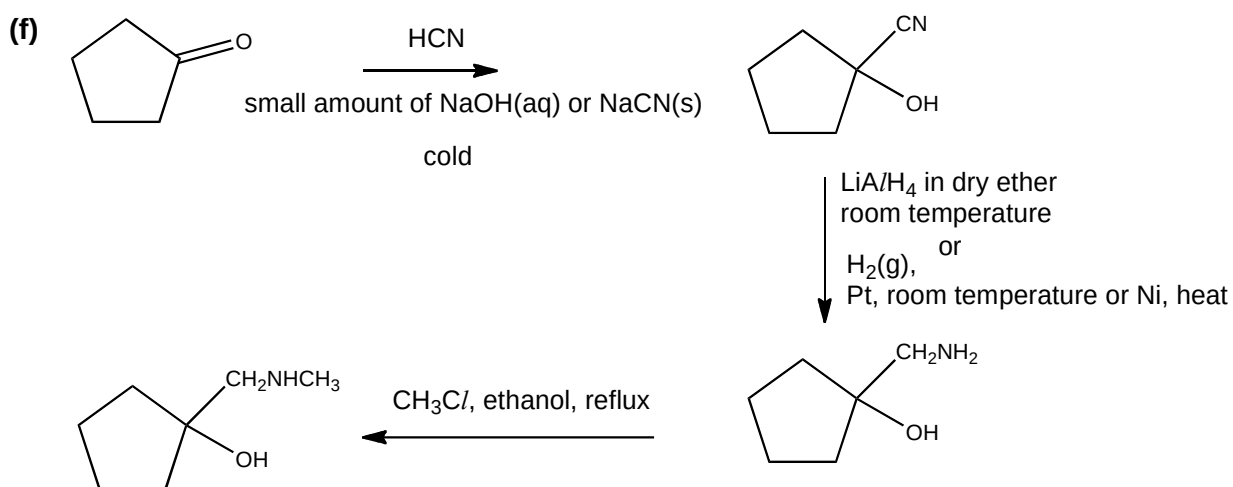
$$\text{No. of moles of } \text{Na}_2\text{CO}_3 = 2.4789 \times 10^{-3} \text{ mol}$$

$$[\text{Na}_2\text{CO}_3] = [(2.4789 \times 10^{-3}) \div 0.02] = \mathbf{0.124 \text{ mol dm}^{-3}}$$

(e)



information	deduction
H produces a yellow precipitate with iodine in alkaline solution.	H undergoes oxidation . H contains CH₃CO- group.
Treatment of H with hot acidified solution of potassium manganate(VII) produces J , C ₅ H ₈ O ₃ , along with a gas that forms white precipitate in limewater.	H undergoes oxidation (or oxidative cleavage) with hot acidified potassium manganate(VII) to produce CO ₂ (g) H contains a terminal alkene
H was also observed to be able to readily decolourise bromine in tetrachloromethane	H undergoes electrophilic addition with Br ₂ in CCl ₄ H contains a alkene



[Total: 20]